

How can I generate a Toeplitz matrix in the correct form for performing discrete convolution?

Asked 7 years, 4 months ago Modified 7 years, 4 months ago Viewed 11k times

Discrete convolution can be performed via the Toeplitz matrix, as shown below ([Wiki article](#)):

$$y = h * x = \begin{bmatrix} h_1 & 0 & \dots & 0 & 0 \\ h_2 & h_1 & \dots & \vdots & \vdots \\ h_3 & h_2 & \dots & 0 & 0 \\ \vdots & h_3 & \dots & h_1 & 0 \\ h_{m-1} & \vdots & \dots & h_2 & h_1 \\ h_m & h_{m-1} & \vdots & \vdots & h_2 \\ 0 & h_m & \dots & h_{m-2} & \vdots \\ 0 & 0 & \dots & h_{m-1} & h_{m-2} \\ \vdots & \vdots & \vdots & h_m & h_{m-1} \\ 0 & 0 & 0 & \dots & h_m \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}$$

Note that this is not the exact same form as as the general Toeplitz matrix, but it has experienced various shifts and zero-paddings.

Is there a way to achieve this in numpy purely based on `roll`, `hstack` etc., i.e. without using any `for` loops? I have tried all sorts of shifts but I can't really get it in to the form shown above.

[python](#) [numpy](#) [linear-algebra](#) [convolution](#) [toeplitz](#) [Edit tags](#)

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edited Dec 30, 2015 at 20:23

asked Dec 30, 2015 at 19:50



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1 Answer

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Yes, you can use [scipy.linalg.toeplitz](#):

```
import numpy as np
from scipy import linalg
```

```
h = np.arange(1, 6)
```



```
padding = np.zeros(h.shape[0] - 1, h.dtype)
first_col = np.r_[h, padding]
first_row = np.r_[h[0], padding]
```

```
H = linalg.toeplitz(first_col, first_row)
```

```
print(repr(H))
# array([[1, 0, 0, 0, 0],
#        [2, 1, 0, 0, 0],
#        [3, 2, 1, 0, 0],
#        [4, 3, 2, 1, 0],
#        [5, 4, 3, 2, 1],
#        [0, 5, 4, 3, 2],
#        [0, 0, 5, 4, 3],
#        [0, 0, 0, 5, 4],
#        [0, 0, 0, 0, 5]])
```

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edited Dec 30, 2015 at 20:22

answered Dec 30, 2015 at 20:02



ali_m

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would this work for 2x2 numpy arrays as input signals ? – [Francisco Vargas](#) Dec 30, 2015 at 20:44



No, `scipy.linalg.toeplitz` treats all of its inputs as 1D arrays. I'm not sure what a 2x2 input array would mean in this context, anyway. – [ali_m](#) Dec 30, 2015 at 20:51



convolution between an image and a an image kernel for example. – [Francisco Vargas](#) Dec 30, 2015 at 20:53



which apparently can be achieved in matlab through this method



stackoverflow.com/questions/16798888/... – [Francisco Vargas](#) Dec 30, 2015 at 20:56

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As in that MATLAB question, if your kernel is x-y separable then you could express it as two 1D vectors of weights, then construct two separate Toeplitz matrices as above and compute two separate dot products along different axes of your image array. – [ali_m](#) Dec 30, 2015 at 20:59 ✎